# Exercise Sheet 9
# Predicate Logic – Natural Deduction & Semantics

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**Note that question 1 is marked as being assessed. Try to also answer questions 2 & 6.**

1. **assessed:** Provide a Natural Deduction proof of

$$(\forall x.\forall y.q(x,y) \to p(x)) \to \forall x.(\exists y.q(x,y)) \to p(x)$$

2. Provide a Natural Deduction proof of $(\exists x.p(x)) \to (\forall x.\forall y.p(x) \to q(x,y)) \to (\exists x.\forall y.q(x,y))$

3. (This is a hard question) Consider the following domain and signature:

   - Functions: $\texttt{pi1}, \texttt{pi2}, \texttt{swap}$ (arity 1); $\texttt{pair}$ (arity 2)
   - Predicates: $=$ (arity 2)

   Consider the following formulas that capture the "definition" of `swap`, and part of the specifications of the other symbols:

   - let $A_1$ be $\forall x.\forall y.\texttt{swap}(\texttt{pair}(x,y)) = \texttt{pair}(y,x)$
   - let $A_2$ be $\forall x.\forall y.x = y \to \texttt{pi1}(x) = \texttt{pi1}(y)$
   - let $A_3$ be $\forall x.\forall y.\texttt{pi1}(\texttt{pair}(x,y)) = x$
   - let $A_4$ be $\forall x.\forall y.\forall z.x = y \to y = z \to x = z$

   The goal is to prove the following property of `swap`, which we call $C$:

$$\forall x.\forall y.\texttt{pi1}(\texttt{swap}(\texttt{pair}(x,y))) = y$$

   i.e., provide a Natural Deduction proof of $A_1 \to A_2 \to A_3 \to A_4 \to C$

4. Consider the following:

   - The signature $\langle\langle \texttt{zero}, \texttt{succ}\rangle, \langle \texttt{even}, \texttt{odd}, >\rangle\rangle$
   - The model $M$: $\langle \mathbb{N}, \langle 0, +1\rangle, \langle\{\langle 0\rangle, \langle 2\rangle, \langle 4\rangle, \dots\}, \{\langle 1\rangle, \langle 3\rangle, \langle 5\rangle, \dots\}, \{\langle 1,0\rangle, \langle 2,0\rangle, \langle 2,1\rangle, \dots\}\rangle\rangle$
   - Formally, because `zero` has arity 0, its interpretation is a function from an empty tuple to 0, written $\langle\rangle \mapsto 0$, which we often write as 0 for simplicity.
   - We write $+1$ for the function that given a number increments it by 1. Formally, this function is $\langle x\rangle \mapsto x{+}1$.
   - The interpretation of `even` can also be written as follows: $\{\langle x\rangle \mid x \text{ is even}\}$.
   - The interpretation of `odd` can also be written as follows: $\{\langle x\rangle \mid x \text{ is odd}\}$.
   - The interpretation of $>$ can also be written as follows: $\{\langle x,y\rangle \mid x > y\}$.

   Prove that $\vDash_{M,\cdot} \forall x.\texttt{even}(x) \to \exists y.\texttt{odd}(y) \land y > x$. Detail your answer as we did in the lectures.

5. Consider the above signature and model $M$. Prove that $\neg \vDash_{M,\cdot} \forall x.\texttt{even}(x) \to \texttt{even}(\texttt{succ}(x))$. Detail your answer as we did in the lectures.

6. Consider the above signature. Provide a model $M'$ such that $\vDash_{M',\cdot} \forall x.\texttt{even}(x) \to \texttt{even}(\texttt{succ}(x))$.